

# Chemistry

## PART – II

### SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

18. What is not produce in given reaction change?  
 Sodium - amalgam  $\xrightarrow{\text{Water}}$   
 (A)  $\text{H}_2$  (B)  $\text{NaOH}$   
 (C)  $\text{Hg}$  (D)  $\text{O}_2$
19. The compound formed by dissolving gold and platinum in aqua-regia is  
 (A)  $[\text{AuCl}_4]^{3-}$  and  $[\text{PtCl}_6]^{2-}$  (B)  $[\text{AuCl}_4]^-$  and  $[\text{PtCl}_6]^{4-}$   
 (C)  $[\text{AuCl}_4]^-$  and  $[\text{PtCl}_6]^{2-}$  (D)  $[\text{AuCl}_4]^-$  and  $[\text{PtCl}_6]^{2-}$
20. What is the emf of a cell containing two hydrogen electrodes, the negative one in contact with  $10^{-6} \text{ M OH}^-$  and the positive one in contact with a  $0.1 \text{ M CH}_3\text{COONH}_4$  solution?  
 Given  $\text{pK}_a (\text{CH}_3\text{COOH}) = 4.74$  and  $\text{pK}_a (\text{NH}_4^+) = 9.26$   
 (A) 0.059 volt (B) 0.59 volt  
 (C) 0.0059 volt (D) 5.9 volt

21.  $\text{PhCONH}_2 + \text{H} \begin{array}{c} \text{Me} \\ | \\ \text{C} \\ | \\ \text{CONH}_2 \end{array} \text{D} \xrightarrow[\Delta]{\text{Br}_2 + \text{KOH}} ?$

Which of the following statement is incorrect regarding the given reaction?

- (A)  $\text{PhNH}_2$  and  $\text{H} \begin{array}{c} \text{Me} \\ | \\ \text{C} \\ | \\ \text{NH}_2 \end{array} \text{D}$  are produced
- (B) Cross-over products  $\text{PhNH}_2$  and  $\text{H} \begin{array}{c} \text{Me} \\ | \\ \text{C} \\ | \\ \text{*NH}_2 \end{array} \text{D}$  are also produced
- (C) Migration of  $-\text{R}$  (alkyl group) of  $\text{R} - \text{CONH}_2$  takes place with retention of configuration.
- (D)  $\text{PhCONHBr}$  and  $\text{H} \begin{array}{c} \text{Me} \\ | \\ \text{C} \\ | \\ \text{CONHBr} \end{array} \text{D}$  are the intermediates formed